Formal Verification of Alternative Mathematical Frameworks: A Case Study Using Lean 4

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Abstract. We present a systematic protocol for applying interactive theorem provers to claims originating from alternative or non-standard mathematical frameworks. Using Lean 4 [**?**] and mathlib [**?**], we extract formalizable assertions from such sources, translate them into dependent type theory, and attempt proof or disproof—reporting results without prejudice. We demonstrate the protocol through a case study on Eric Dollard's "versor algebra," a non-standard extension of electrical engineering mathematics. Results are deliberately mixed: we verify a $\mathbb{Z}_4$ cyclic group structure (trivially correct), disprove a claimed versor-form equivalence (sign error), establish by algebraic necessity that the proposed operators collapse to standard 4th roots of unity, and trace the algebraic path from $\mathbb{Z}_4$ through Clifford algebras $\mathrm{Cl}(1,1)$, $\mathrm{Cl}(3,0)$, $\mathrm{Cl}(1,3)$ to spacetime algebra. We propose a five-category taxonomy for classifying claims from alternative frameworks: Verified, Disproved, Algebraic Necessity, Ambiguous, and Unfalsifiable. To our knowledge, this represents the first application of interactive theorem provers to formalize claims from non-mainstream mathematical sources.

Keywords: formal verification · Lean 4 · alternative mathematics · Clifford algebras · versor algebra

1 Introduction

Interactive theorem provers have verified some of mathematics' most challenging results: Hales' Kepler conjecture via the Flyspeck project [**?**], the Four Color Theorem [**?**], Scholze's liquid tensor experiment [**?**], the polynomial Freiman-Ruzsa conjecture [**?**], and the consistency of Quine's New Foundations [**?**]. In each case, the target was mainstream mathematics—contested, complex, or uncertain, but arising within accepted mathematical frameworks.

A conspicuous gap remains: no prior work applies interactive theorem provers to claims from *alternative* or *non-standard* mathematical frameworks. By "alternative framework" we mean a coherent system of mathematical claims originating outside peer-reviewed mathematics, typically from independent researchers, that proposes non-standard definitions, operations, or equivalences. Such frameworks are distinct from non-standard *foundations* (e.g., HoTT, constructive

mathematics), which are rigorous alternative approaches proposed within the mathematical community [**?**].

This gap is structural, not accidental. Four factors conspire to maintain it:

1. **Cultural separation.** Formal verification and alternative mathematics communities have zero overlap.
2. **No academic incentive.** Formalizing alternative claims yields no new theorems and no library contributions, while risking the appearance of legitimizing pseudoscience.
3. **Asymmetric difficulty.** Translating informal non-standard claims into dependent type theory requires both mathematical expertise and judgment about ambiguous notation.
4. **No client.** Proponents of alternative frameworks rarely seek formal verification (which might disprove claims), and mainstream mathematicians see no need (they already reject such claims on conventional grounds).

We argue that this gap represents a missed opportunity. Formal verification of alternative frameworks is methodologically interesting precisely when it produces *mixed* results: some claims verified, others disproved, still others classified as ambiguous or unfalsifiable. Pure debunking is uninteresting (it offers no methodological contribution), and pure verification is unlikely for frameworks that diverge from standard mathematics. Mixed results demonstrate the discriminating power of formal methods and reveal genuine mathematical structure worth analyzing.

Contributions. This paper makes four contributions:

1. A six-step *protocol* for extracting, translating, and verifying claims from alternative mathematical frameworks (**??**).
2. A *case study* applying the protocol to Eric Dollard's versor algebra, producing over 200 formally verified theorems across 8 Lean 4 proof files with zero `sorry` gaps (**??**); the full repository extends to 59 proof files with over 2,537 verified declarations, including four Lie algebras with certified mathlib `LieRing`/`LieAlgebra` instances.
3. A five-category *taxonomy* for classifying claims from alternative frameworks: Verified, Disproved, Algebraic Necessity, Ambiguous, and Unfalsifiable (**??**).
4. A constructive *algebraic hierarchy* tracing the path from the source claims through Clifford algebras to spacetime algebra, demonstrating that formal methods can suggest corrections, not merely judge (**??**).

2 Related Work

2.1 Formalization of Contested Mathematics

The closest parallels to our work involve formalizations where the mathematical community itself had doubts about correctness. The Flyspeck project [**?**] verified Hales' proof of the Kepler conjecture after referees could not fully evaluate

it. The liquid tensor experiment [**?**] verified a theorem where even the author (Scholze) had reservations. Wilshaw's formalization of NF consistency [**?**] addressed a proof that mainstream reviewers could not confidently evaluate. In each case, the target was contested-within-mainstream, not alternative/fringe.

2.2 Formalization of Physical Theories

Tooby-Smith's PhysLean/HepLean project [**?**] digitalizes mainstream physics in Lean 4, from classical mechanics through quantum field theory. Bobbin et al. [**?**] formalized Langmuir and BET adsorption theories, demonstrating that Lean catches faulty logic and reveals hidden assumptions—the closest methodological parallel to our work, though applied to mainstream physical chemistry.

2.3 Historical Mathematics Formalization

Fleuriot [**?**] formalized Newton's geometric proofs in Isabelle/HOL and discovered a flaw in one of Newton's proofs. This is the closest near-miss to our work: a proof assistant applied to historical non-rigorous reasoning that finds errors. The distinction is that Newton is mainstream; the "non-standard" aspect is proof style, not framework.

2.4 Clifford Algebra Formalization

Wieser and Song [**?**] formalized Clifford algebras in Lean 3/mathlib, providing the mathematical objects that overlap with our case study. Their work formalizes standard theory; ours applies it to non-standard claims.

2.5 Philosophy of Formal Verification

Avigad [**?**] examines how formal verification contributes to mathematical understanding. Imbert and Ardourel [**?**] analyze formal verification's epistemological role in computational science. Shulman [**?**] discusses proof assistants for exploring alternative foundations (HoTT), though these are rigorous alternatives proposed by credentialed mathematicians.

3 The Protocol

We propose a six-step protocol for applying interactive theorem provers to alternative mathematical frameworks. The individual techniques at each step are standard; the contribution is the systematic application to a novel domain.

1. **Source Identification.** Identify an alternative mathematical framework containing specific algebraic or structural claims amenable to formalization. The framework must make assertions beyond rhetorical or philosophical claims.

2. **Claim Extraction.** Enumerate all formalizable assertions from the source material. Separate mathematical claims ("$j^4 = 1$") from rhetoric ("electricity is fundamental"), physics claims ("aether exists"), and philosophical framing. Document the extraction process, including claims deliberately excluded and why.
3. **Translation.** Map each extracted claim to dependent type theory. Document all translation choices, especially where source notation is ambiguous. When multiple interpretations exist, formalize the most charitable one (*principle of charity*).
4. **Verification Attempt.** For each translated claim, attempt proof or disproof using standard tactics. Record the outcome: proved, disproved (with counterexample or contradiction), or neither (claim may be ambiguous or unformalizable).
5. **Taxonomy Classification.** Assign each claim to one of five categories (**??**).
6. **Neutral Reporting.** Present all results without editorial judgment on the source framework. The role of the formal verifier is analogous to that of an auditor: to report findings, not to advocate or condemn.

4 Case Study: Dollard's Versor Algebra

4.1 Background

Eric Dollard is an independent researcher in electrical engineering who has proposed a "versor algebra" as an extension of standard complex-number methods for analyzing polyphase power systems [**?**] and transmission lines [**?**,**?**]. The framework introduces three operators: j (the standard imaginary unit), h (a proposed "hyperbolic" operator with $h^2 = 1$, $h^1 = -1$), and $k = hj$ (a "sequence" operator). Dollard claims these operators form a novel algebraic structure with applications to electromagnetic theory.

The source materials are four self-published documents: *Lone Pine Writings*, *Four Quadrant Theory*, *Versor Algebra*, and *Advanced Versor Algebra II-23* [**?**,**?**]. These documents are not peer-reviewed and are not widely available; we therefore reproduce the core definitions here for self-containment. We extracted 17 formalizable claims from these sources.

Dollard's axioms. Dollard postulates three operators j, h, k in a unital associative algebra, subject to five stated properties:

(A1) $j^2 = -1$ (the standard imaginary unit),
(A2) $hj = k$ (product definition of the sequence operator),
(A3) $jk = 1$ (cancellation),
(A4) $h^1 = -1$ (h evaluates to -1),
(A5) $h^2 = 1$ (h squared is unity).

Dollard writes $h = \sqrt{+1}^{1/2}$, using both the square root and a fractional exponent, which is mathematically ambiguous (see **??**). The intended interpretation appears to be an operator that squares to $+1$ but is itself -1—distinct from the principal square root.

Sign convention. A critical detail: Dollard uses the admittance convention $Y = G - jB$ (negative susceptance), not the standard $Y = G + jB$. This affects the telegraph equation expansion. Our formalization verifies both conventions for comparison.

4.2 Verified: $\mathbb{Z}_4$ Cyclic Group Structure

Dollard's operators $\{1, j, h, k\}$ with $j^2 = -1$, $h = -1$, $k = hj = -j$, and $jk = 1$ form a group under multiplication. This group is isomorphic to $\mathbb{Z}_4$, the cyclic group of order 4, which is also the group of 4th roots of unity $\{1, i, -1, -i\}$ under complex multiplication.

Listing 1.1. Versor operators as 4th roots of unity (excerpt from `basic_operators.lean`)

```
def j : ℂ:= Complex.I
def h : ℂ:= -1
def k : ℂ:= h * j

theorem j_fourth_power : j ^ 4 = 1 := by ...
theorem h_squared : h ^ 2 = 1 := by ...
theorem jk_cancellation : j * k = 1 := by ...
```

All 16 products of the Cayley table are verified. The structure is trivially correct—it is standard complex arithmetic. The verification confirms Dollard's framework includes correct standard results, though the algebraic content is not novel.

4.3 Disproved: Versor Form Equivalence

Dollard claims that the telegraph equation product $ZY = (R + jX)(G - jB)$ can be equivalently written in "versor form" as:

$$ZY = h(XB + RG) + j(XG - RB)$$

With $h = -1$ (as established by the $\mathbb{Z}_4$ structure):

$$\begin{aligned} \text{Versor:} \quad & h(XB{+}RG) + j(XG{-}RB) = -(XB{+}RG) + j(XG{-}RB) \\ \text{Telegraph:} \quad & (RG{+}XB) + j(XG{-}RB) \end{aligned}$$

The imaginary parts match, but the real parts differ by sign. The forms are equal only when $RG + XB = 0$—a degenerate case where both dissipation and storage vanish simultaneously.

Listing 1.2. Versor form disproof (excerpt from `telegraph_equation.lean`)

```
theorem versor_form_value (R X G B : ℝ) :
    h * (↑(factor_XB X B + factor_RG R G) : ℂ) +
    j * (↑(factor_XG X G - factor_RB R B) : ℂ) =
```

```
  Complex.mk (-(factor_XB X B + factor_RG R G))
             (factor_XG X G - factor_RB R B) := by
 simp only [h, j]
 apply Complex.ext <;> simp [...]
```

This disproof is robust: it holds regardless of sign convention (Dollard or standard), because $h = -1$ negates the real part in either case. Furthermore, the versor form is *irreparable*: we prove that the coefficient on the real part must equal 1 for equivalence (`versor_repaired_with_one`), but $h \neq 1$ (`h_ne_one`).

4.4 Algebraic Necessity: $h = -1$ Forced

The most structurally interesting result: Dollard's three core axioms $j^2 = -1$, $hj = k$, $jk = 1$ *alone* force $h = -1$ over any field. The proof is elementary:

$$jk = 1 \implies k = j^{-1} = -j, \qquad hj = k = -j \implies h = -1$$

The axioms $h^1 = -1$ and $h^2 = 1$ are redundant—they follow from the three core axioms. This means Dollard's algebra has zero degrees of freedom: any unital associative algebra satisfying the core axioms is forced to $\mathbb{Z}_4$.

Listing 1.3. Algebraic necessity over any field (excerpt from `algebraic_necessity.lean`)

```
-- Section variables: {F : Type*} [Field F] {j' h' k' : F}
theorem algebraic_necessity_general
    (hj2 : j' ^ 2 = -1) (hhj : h' * j' = k')
    (hjk : j' * k' = 1) :
    h' = -1 := by ...
```

This result closes the "rehabilitation question": to obtain a non-trivial h (e.g., in $\mathrm{Cl}(1,1)$ where $e_1^2 = 1$ but $e_1 \neq \pm 1$), one must drop at least two of Dollard's three core axioms. The resulting algebra is a *different* system, not a repair of Dollard's.

The D_8 alternative. Wilson [**?**] observes that versors in the standard Clifford-algebraic sense satisfy anticommutativity: in $\mathrm{Cl}(1,1)$, generators e_1, e_2 with $e_1^2 = +1$, $e_2^2 = -1$ satisfy $e_1 e_2 = -e_2 e_1$. Identifying $h = e_1$ and $j = e_2$, the group generated by $\{j, h\}$ has 8 elements $\{\pm 1, \pm j, \pm h, \pm jh\}$ and is isomorphic to the dihedral group D_8. Our algebraic necessity result confirms this from the complementary direction: Dollard's (implicit) commutativity assumption forces $h = -1$, collapsing D_8 to $\mathbb{Z}_4$. The sign error identified in **??** is a direct consequence of this collapse—had Dollard worked in $\mathrm{Cl}(1,1)$ with anticommuting generators, the versor form would require different algebraic treatment entirely.

4.5 Ambiguous: Notation Issues

Dollard writes $h = \sqrt{+1}^{1/2}$, which is mathematically ambiguous. In $\mathbb{R}$, $\sqrt{1} = 1$ and $1^{1/2} = 1$, so $h = 1$—but Dollard claims $h = -1$. The notation admits multiple interpretations, none of which are standard.

Formal methods precisely *identify* this ambiguity as a contribution: rather than resolving it by fiat, we show that the algebraic necessity result (**??**) makes the notation irrelevant. Whatever $\sqrt{+1}^{1/2}$ means, h must equal -1 in any algebra satisfying Dollard's core axioms.

4.6 Unfalsifiable: Physics Claims

Dollard defines Ψ (total dielectric induction, in Coulombs) and Φ (total magnetic induction, in Webers), with $Q = \Psi \cdot \Phi$ and $W = dQ/dt$. Dimensionally, Coulomb $\times$ Weber $=$ J $\cdot$ s (action), making Q the action and W the energy. This is confirmed by primary source analysis: Dollard's $d\Psi/dt = I$ (current) and $d\Phi/dt = E$ (voltage) exactly reproduce the Lagrangian circuit variables where charge q is the generalized coordinate and flux λ is the conjugate momentum. The dimensional analysis is verified in Lean 4 (`circuit_action.lean`, 0 `sorry`). The framework is dimensionally correct but not novel—it is Lagrangian circuit theory, known since Cherry [**?**].

Several remaining claims—"electricity is fundamental," "aether exists as a fifth state of matter," "$W = dQ/dt$ replaces $E = mc^2$"—are not addressable by formal mathematical methods. We classify these as Unfalsifiable (by theorem provers), noting that this classification is itself a useful result: it separates the checkable from the uncheckable.

5 A Taxonomy for Claims from Alternative Frameworks

Based on the case study, we propose a five-category taxonomy for classifying claims from alternative mathematical frameworks:

Table 1. Five-category taxonomy for claims from alternative frameworks

Category	Definition	Example
Verified	Claim is mathematically correct; formal proof produced	$\mathbb{Z}_4$ group structure
Disproved	Claim is mathematically false; formal counterexample or contradiction produced	Versor form equivalence
Algebraic Necessity	Premises force a conclusion the author did not intend	$h = -1$ forced by core axioms
Ambiguous	Claim cannot be formalized without resolving notation/definition gaps	$h = \sqrt{+1}^{1/2}$
Unfalsifiable	Claim is not addressable by formal methods	Physics/empirical claims

We argue that mixed results across these categories are the *expected* and methodologically interesting case. Uniform rejection would suggest the frame-

work is too incoherent to formalize. Uniform verification would suggest the framework is merely standard mathematics with non-standard notation. Mixed results—some verified, some disproved, some ambiguous—reveal genuine structure worth analyzing and demonstrate the discriminating power of formal methods.

The Algebraic Necessity category merits special attention. It captures cases where the author's axioms entail consequences they did not recognize or intend. This phenomenon has precedents across several fields: as a consequence of Hilbert's implicit definitions (the Frege-Hilbert controversy), as "implicit commitment" in epistemic logic [**?**], and as overspecification in formal methods. The Banach-Tarski paradox is a canonical example at larger scale—the axiom of choice forces consequences its adopters did not foresee.

We note a duality with Lakatos's *Proofs and Refutations* [**?**]: Lakatos addresses axiom systems that are too *weak*, admitting unintended models ("monsters" requiring barring). Our Algebraic Necessity addresses axiom systems that are too *strong*, forcing unintended consequences. These are dual failure modes:

Failure	**Direction**	**Lakatos**	**This work**
Axioms too weak	Admit unintended models	Monster-barring	—
Axioms too strong	Force unintended conclusions	—	Alg. Necessity

This duality suggests that Algebraic Necessity fills a genuine gap in the vocabulary for analyzing axiom systems—particularly relevant when formal methods are applied to frameworks whose authors may not have fully explored the consequences of their axioms.

6 The Clifford Algebra Hierarchy

Beyond classification, formal methods can suggest *corrections*. We trace the algebraic path from Dollard's $\mathbb{Z}_4$ to the spacetime algebra where electromagnetism becomes geometry:

Table 2. Algebraic hierarchy from $\mathbb{Z}_4$ to spacetime algebra

Level	**Algebra**	**Dim**	**Physical Content**
1	$\mathbb{Z}_4$	4	Versor operators (trivial, forced by axioms)
2	$\mathrm{Cl}(1,1)$	4	Idempotent wave decomposition
3	$\mathrm{Cl}(3,0)$	8	Pauli algebra; $F = E + IB$
4	$\mathrm{Cl}(1,3)$	16	Spacetime algebra; $\nabla F = J$

Each level is formalized in Lean 4 with zero `sorry` gaps.

At Level 2, $\mathrm{Cl}(1,1)$ provides what Dollard was reaching for: a non-trivial operator e_1 with $e_1^2 = +1$ but $e_1 \neq \pm 1$. However, this algebra is non-commutative

($e_1e_2 \neq e_2e_1$), and Dollard's axiom $jk = 1$ fails ($e_2 \cdot e_{12} = e_1 \neq 1$). The algebra offers idempotent projectors $P_\pm = \frac{1 \pm e_1}{2}$ that decompose signals into forward and backward wave components—a genuine mathematical tool for transmission line analysis.

At Level 4, $\mathrm{Cl}(1,3)$ is the spacetime algebra [**?**,**?**] where Maxwell's four equations collapse to one: $\nabla F = J$. This is the genuine unified field framework for classical electromagnetism, achieved through geometric algebra rather than through Dollard's commutative $\mathbb{Z}_4$.

The hierarchy demonstrates that Dollard's *instinct*—algebraic unification of electromagnetic phenomena—was correct, while his *execution*—a commutative group with four elements—was far too restrictive. Formal methods can articulate this distinction precisely.

7 Discussion

7.1 Why Mixed Results Matter

The case study produced all five taxonomy categories across 17 claims. This distribution validates the protocol's discriminating power: it is neither a rubber stamp nor a blanket dismissal.

The sign error in the versor form (**??**) could not have been established with certainty without formalization. Informal mathematical reasoning might notice the discrepancy, but Lean provides machine-checked certainty—and proves the error is irreparable.

7.2 Why the Gap Exists

The four structural factors identified in **??**—cultural separation, no academic incentive, asymmetric difficulty, no client—explain why this application domain has been overlooked. We note that none of these factors are *mathematical*; the gap is sociological. The mathematical tools are entirely adequate: standard Lean 4 tactics suffice for all proofs in our case study, and no custom tactics or library extensions were required.

7.3 The Constructive Dimension

Our Clifford algebra hierarchy (**??**) goes beyond demarcation to *mathematical rehabilitation*: it traces the path from the alternative claim to the correct mathematical framework. This constructive contribution— showing not just what is wrong but what the author *should* have said—may be the most valuable aspect of applying formal methods to alternative frameworks.

7.4 Limitations

Our protocol is demonstrated on a single case study; generalizability to other alternative frameworks is conjectured but not proven. Dollard's source materials are self-published presentations, not peer-reviewed papers, introducing translation ambiguities that are inherent to the domain. All Lean 4 proofs compile against mathlib (the full 59-file repository builds with zero errors, zero `sorry` gaps). We formalize algebraic and structural claims only; we make no claims about the physics, engineering utility, or broader validity of Dollard's work.

8 Conclusion

We have demonstrated that interactive theorem provers can be productively applied to claims from alternative mathematical frameworks. The protocol produces mixed results—verified, disproved, algebraically necessary, ambiguous, and unfalsifiable—that reveal genuine mathematical structure. The five-category taxonomy provides a systematic classification scheme. The Clifford algebra hierarchy shows that formal methods can constructively suggest the "correct" version of an alternative framework's aspirations.

The value of this approach lies not in judgment but in discrimination: formal methods separate what is mathematically checkable from what is not, what is correct from what is wrong, and what is novel from what is standard. We call for further case studies on other alternative mathematical frameworks to validate and refine the protocol.

The case study presented here covers the 8 proof files most directly tied to the Dollard verification. The full repository extends the algebraic hierarchy through grand unified theories (SO(10), SO(14)) and exceptional algebras (E_8), totaling 59 proof files with over 2,537 verified declarations—all compiling with zero `sorry` gaps. Recent work includes mathlib `LieRing` and `LieAlgebra` $\mathbb{R}$ typeclass instances for four Lie algebras (so(1,3), sl(3), sl(5), so(10)), connecting the hand-built structures to mathlib's abstract type system, and a machine-verified resolution of the three-generation problem via $E_8 \supset \mathrm{SU}(9)/\mathbb{Z}_3$. Source code, source materials, and analysis documents are available from the corresponding author upon reasonable request.

Appendix: Complete Claim Summary

Note on mathlib. Mathlib includes a comprehensive formalization of Clifford algebras (`Mathlib.LinearAlgebra.CliffordAlgebra`) [**?**] with grading, spin groups, and conjugation. Our case study uses explicit finite-dimensional representations for pedagogical clarity, but a natural extension would unify our ad-hoc $\mathrm{Cl}(p, q)$ types with mathlib's abstract `CliffordAlgebra` via isomorphism proofs. As a step in this direction, four Lie algebras in the extended repository—so(1,3), sl(3), sl(5), and so(10)—now carry certified `LieRing` and `LieAlgebra` $\mathbb{R}$ instances via `Mathlib.Algebra.Lie.Basic`, connecting the explicit constructions to mathlib's abstract type system.

Table 3. All 17 formalized claims with taxonomy classification

#	Claim	Category	Proof File
1	$\{1, j, h, k\}$ as 4th roots of unity	VERIFIED	`basic_operators`
2	Telegraph expansion $(R + jX)(G - jB)$	VERIFIED	`telegraph_equation`
3	Four-factor independence	DISPROVED*	`telegraph_equation`
4	Versor form equivalence	DISPROVED	`telegraph_equation`
5	Universal polyphase formula	VERIFIED	`polyphase_formula`
6	$h = \sqrt{+1}^{1/2}$ notation	AMBIGUOUS	`algebraic_necessity`
7	Sequence periodicity	VERIFIED	`basic_operators`
8	$Q = \Psi \cdot \Phi$ (action)	VERIFIED†	`circuit_action`
9	Electricity as fundamental	UNFALSIFIABLE	—
10	Aether as 5th state	UNFALSIFIABLE	—
11	$W = dQ/dt$ replaces $E = mc^2$	UNFALSIFIABLE	—
12	Over-unity energy	UNFALSIFIABLE	—
13	Dielectric counterspace	UNFALSIFIABLE	—
14	Quaternary expansion	VERIFIED‡	(computational)
15	Fortescue is domain-specific	VERIFIED	(empirical)
16	A_4 matches versor matrix	VERIFIED	(computational)
17	$jk = 1$ forces $h = -1$	ALG. NECESSITY	`algebraic_necessity`

*Four terms exist but satisfy $(RG)(XB) = (XG)(RB)$; not independent.
†Dimensionally correct (Weber·Coulomb = J·s) but not novel (Cherry, 1951).
‡Mathematically correct but practically useless (deprecated).

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References

1. Hales, T., et al.: A formal proof of the Kepler conjecture. Forum of Mathematics, Pi **5**, e2 (2017)
2. Gonthier, G.: Formal proof—The Four-Color Theorem. Notices of the AMS **55**(11), 1382–1393 (2008)
3. Commelin, J., et al.: Completion of the Liquid Tensor Experiment. Lean Community (2022)
4. Tao, T., et al.: Polynomial Freiman-Ruzsa conjecture. Lean 4 formalization (2023)
5. Wilshaw, S.: New Foundations is consistent. Lean formalization, leanprover-community/con-nf (2024)

6. Shulman, M.: Strange new universes: Proof assistants and synthetic foundations. Bull. AMS **61**(2) (2024)
7. Tooby-Smith, J.: HepLean: Digitalising high energy physics. Computer Physics Communications **308** (2025)
8. Bobbin, M.P., et al.: Formalizing chemical physics using the Lean theorem prover. Digital Discovery, RSC (2024)
9. Fleuriot, J.: A Combination of Geometry Theorem Proving and Nonstandard Analysis with Application to Newton's Principia. Springer (2001)
10. Wieser, E., Song, U.: Formalizing Geometric Algebra in Lean. Advances in Applied Clifford Algebras **31** (2021)
11. Avigad, J.: Understanding, formal verification, and the philosophy of mathematics. Journal of the Indian Council of Philosophical Research (2021)
12. Imbert, C., Ardourel, V.: Formal verification, scientific code, and epistemological heterogeneity. Philosophy of Science (2023)
13. de Moura, L., Ullrich, S.: The Lean 4 theorem prover and programming language. CADE-28, LNCS **12699** (2021)
14. The mathlib community: The Lean mathematical library. CPP 2020.
15. Fortescue, C.L.: Method of symmetrical co-ordinates applied to the solution of polyphase networks. AIEE Transactions **37**(2), 1027–1140 (1918)
16. Hestenes, D.: Space-Time Algebra. Gordon and Breach (1966)
17. Doran, C., Lasenby, A.: Geometric Algebra for Physicists. Cambridge University Press (2003)
18. Dollard, E.P.: Versor Algebra: As Applied to Polyphase Power Systems. Self-published presentation.
19. Dollard, E.P.: Four Quadrant Theory: Advanced Electrical Applications. Self-published presentation.
20. Cherry, E.C.: Some general theorems for non-linear systems possessing reactance. The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science **42**(333), 1161–1177 (1951)
21. Lakatos, I.: Proofs and Refutations: The Logic of Mathematical Discovery. Cambridge University Press (1976)
22. Fagin, R., Halpern, J.Y.: Belief, awareness, and limited reasoning. Artificial Intelligence **34**(1), 39–76 (1988)
23. Wilson, R.A.: Private communication (2026)